

DOM Selectors Practice Notes (JavaScript)

1. Base HTML Structure (Practice Setup)

```
<div id="app">

  <header class="navbar main-nav">
    <h1 class="logo">MyShop</h1>
    <ul class="menu">
      <li class="menu-item active">Home</li>
      <li class="menu-item">Products</li>
      <li class="menu-item">Contact</li>
    </ul>
  </header>

  <section class="products" data-section="products">

    <div class="card featured" data-id="101">
      <h2 class="title">iPhone</h2>
      <p class="price">$1000</p>
      <button class="btn buy-btn">Buy</button>
    </div>

    <div class="card" data-id="102">
      <h2 class="title">Samsung</h2>
      <p class="price">$800</p>
      <button class="btn buy-btn">Buy</button>
    </div>

    <div class="card sale" data-id="103">
      <h2 class="title">OnePlus</h2>
      <p class="price">$700</p>
      <button class="btn buy-btn special">Buy</button>
    </div>

  </section>

</div>
```

2. Basic Selectors

```
document.querySelector(".card")           // first card
document.querySelectorAll(".card")        // all cards
document.querySelector("#app")            // id
document.querySelector("header")          // element
```

3. Combining Selectors

```
document.querySelector(".card.featured") // multiple class
document.querySelector("div.card.featured") // element + class
document.querySelector("[data-id='101']") // attribute
```

4. Descendant vs Child

```
// Descendant (any level inside)
document.querySelectorAll(".products .title")

// Direct child only
document.querySelectorAll(".products > .card")
```

- ✓ Space " " → inside (any level)
 - ✓ > → direct child only
-

5. Sibling Selectors

```
document.querySelector(".featured + .card") // next sibling
document.querySelectorAll(".featured ~ .card") // all next siblings
```

6. Attribute Selectors

```
[data-id] // has attribute
[data-id="102"] // exact match
[data-id^="10"] // starts with
[data-id$="3"] // ends with
[data-id*="0"] // contains
```

- ✓ Always use [] for attributes
-

7. Pseudo Classes

```
.menu li:first-child
.card:last-child
.card:nth-child(2)
.card:not(.sale)
```

8. querySelector vs querySelectorAll

```
document.querySelector(".card") // single
document.querySelectorAll(".card") // multiple
```

✓ Use querySelector when only one element expected

9. Extracting Data (IMPORTANT)

```
// get all titles
document.querySelectorAll(".title")
  .forEach(e => console.log(e.innerText))

// store in array
let titles = [...document.querySelectorAll(".title")]
  .map(e => e.innerText)
```

- ✓ Use innerText for text
 - ✗ Avoid innerHTML unless needed
-

10. Same Element → Multiple Ways

Example: OnePlus title

```
document.querySelector(".card.sale .title")
document.querySelector("[data-id='103'] .title")
document.querySelectorAll(".card")[2].querySelector(".title")
```

11. Important Concepts

- ✓ Space vs No Space

```
.card .title // inside
.card.title // same element
```

✓ > VS :first-child

```
.menu > li // all direct children
.menu li:first-child // only first
```

✓ Attribute Rules

```
[data-id="102"] // correct
#app // id selector
.card // class selector
```

✗ [#id] invalid

✗ [data-id#="1"] invalid

12. Best Practices

- ✓ Use simple selectors
 - ✓ Prefer `data-*` attributes
 - ✓ Avoid long chains
 - ✓ Try multiple ways
-

13. Practice Tasks

Task 1

```
document.querySelectorAll(".card")
```

Task 2

```
document.querySelector(".card.featured")
```

Task 3

```
document.querySelectorAll(".menu li")
```

Task 4

```
document.querySelector(".card.sale")
```

Task 5

```
document.querySelector(".card.sale .buy-btn")
```

Task 6

```
document.querySelector(".buy-btn.special")
```

Task 7

```
document.querySelectorAll(".products .title")
```

Task 8

```
document.querySelectorAll(".products > .card")
```

Task 9

```
document.querySelector("[data-id='102']")
```

Task 10

```
document.querySelectorAll("[data-id^='10']")
```

Task 11

```
document.querySelector(".menu li:first-child")
```

Task 12

```
document.querySelector(".card:last-child")
```

Task 13

```
document.querySelectorAll(".card:not(.sale)")
```

Final Goal

🔑 When you inspect any website, think:

- Can I use class?
- Can I use parent?
- Can I use attribute?
- Can I use position?
- Can I combine selectors?

✓ Always try 2–3 ways for same element

Summary

- . → class
 - # → id
 - [] → attribute
 - " " → inside
 - > → direct child
 - : → pseudo
-

💧 Practice in:

- Browser Console (F12)
 - Your own HTML file
 - Real websites
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